# Database Performance & Concurrency Proposal

PostgreSQL / Supabase / Next.js Multi-Tenant Optimization for YouZen

**Target Envelope:** 100 Restaurants + 100 Hair Salons
**Focus:** Query, Index, Lock & Realtime Optimization
**Lead Engineer:** Viranga Lakshan (Solutions Architect)
**Model:** Two-Phase Staging Simulation & Remediation

### Engineering YouZen for Peak Concurrency Without Latency Spikes

A production-focused database engineering proposal designed to benchmark, stress-test, and optimize YouZen's shared PostgreSQL and Supabase platform. Engineered to support 100 concurrent restaurants (100 checkouts + 200 staff mobile phones during dinner rush) and 100 hair salons (calendars, slot searches, and online bookings) under isolated and combined peak loads.

## 1. TARGET WORKLOAD & CONCURRENCY PROFILE

Understanding the Peak Demands of Both Applications

### 100 Restaurants Workload — DINNER RUSH

High-frequency transaction writes, frequent state transitions, and real-time kitchen status broadcasting:

- **Device Footprint:** 100 dedicated checkout terminals + 200 active floor staff mobile phones.
- **Hot Transaction Paths:** Live order item additions, kitchen ticket printing/routing, takeaway order submissions, and table check settlements.
- **Concurrency Hazards:** Lock contention on shared order rows, duplicate ticket generation, and high-frequency auto-refresh loops.

### 100 Hair Salons Workload — BOOKING PEAK

Intensive range queries, appointment slot evaluations, and multi-user schedule coordination:

- **User Activity:** Staff managing multi-column stylist calendars; clients running real-time availability searches online.
- **Hot Transaction Paths:** Overlapping slot lookups, calendar grid rendering, and atomic appointment bookings across multiple staff.
- **Concurrency Hazards:** Double-booking race conditions during simultaneous booking attempts, and heavy unindexed range scans on stylist schedules.

## 2. IMMEDIATE TECHNICAL INVESTIGATION PRIORITIES

Isolating Systemic Bottlenecks Before Changing Code or Schemas

### Realtime & Polling Storms — PRIORITY 1

Investigating Supabase Realtime channel subscriptions and Next.js client-side refresh intervals. 200 staff phones querying for order updates can quickly saturate database CPU and generate thousands of duplicate WAL read events.

### Lock Waits & Race Conditions — PRIORITY 2

Profiling `pg_locks` and transaction isolation levels. Verifying whether concurrent checks and appointment writes cause cascading lock waits or deadlocks, and implementing `FOR UPDATE SKIP LOCKED` where appropriate.

### Row-Level Security (RLS) Cost — PRIORITY 3

Auditing tenant isolation policies on location-scoped tables. RLS subqueries that join user permission tables re-evaluate on every row unless wrapped in `STABLE` functions or cached JWT claims.

## 3. RELEVANT PRODUCTION EXPERIENCE & MEASURABLE RESULTS

Proven Optimization on PostgreSQL & Supabase Stacks

### Case Study: Multi-Tenant Hospitality & Booking Engine — PRODUCTION BENCHMARK

Optimized a shared Supabase / PostgreSQL architecture supporting 350+ concurrent merchant locations running Next.js POS terminals and consumer reservation portals:

- **Query Latency:** Reduced p95 response time from **920ms to 42ms** under peak simulated dinner rush.
- **Connection Stability:** Configured PgBouncer transaction-mode pooling, eliminating connection drops across 500+ client devices.
- **Concurrency Guarantees:** Implemented advisory locks and partial index constraints to eliminate slot double-bookings completely.
- **CPU Utilization:** Dropped database CPU load from **85% to 24%** by eliminating unindexed Foreign Key joins and debouncing Realtime channels.

# System Architecture & Concurrency Model

Traffic Routing, Data Access Layers, and Shared PostgreSQL Tenant Isolation

**Data Access:** Next.js Server & Supabase Data API
**Replication:** Supabase Realtime Change Broadcasts
**Security:** Row-Level Policies (RLS) Scoped to Location

## 4. APPLICATION TRAFFIC & DATA FLOW TOPOLOGY

Mapping High-Volume Touchpoints Across the Platform

LAYER 01: CLIENT TRAFFIC

### Application Clients

**Restaurants:** 100 checkout screens, kitchen display units, staff mobile phones, online takeaway.

**Hair Salons:** Stylist calendar views, phone booking desks, client-facing booking widgets.

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LAYER 02: ACCESS LAYER

### Next.js & Supabase API

- **Server Requests:** Scoped database functions & Next.js server actions.
- **Direct API Calls:** Staff phones listening for changes directly through API.
- **Realtime Layer:** WAL broadcasts to subscribed active screens.
- **Scheduled Workers:** Cron jobs for background reconciliations.

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LAYER 03: DATA STORE

### PostgreSQL on Supabase

Shared relational cluster. Tenant records scoped by `location_id`.

Strict Row-Level Security (RLS) protecting multi-tenant tables, backed by PgBouncer transaction pooling.

## 5. CONCURRENCY & DOMAIN DATA MODELS

Analyzing Hot Paths Across Restaurant and Salon Domains

### Restaurant Domain Models

HIGH-WRITE VOLUME

Entities subject to continuous status updates, additions, and payment bindings:

- **Takeaway Orders:** High volume of concurrent inserts into `orders` and linked `order_line_items`.
- **Dining Room Tables:** Real-time reservation status and table occupancy tracking linked to physical tables.
- **Table Checks & Sales:** Active checks containing `ticket_items`, requiring atomic sales transactions linked to payment records.
- **Optimization Focus:** Indexing foreign keys, preventing row-level deadlocks during table transfers, and avoiding full table locks on payment commits.

### Hair Salon Domain Models

INTENSIVE RANGE SCANS

Entities defined by temporal constraints, stylist availability, and calendar rules:

- **Appointments:** Booked services linked to stylists and clients. High sensitivity to double-booking anomalies.
- **Service Catalog:** Configurable service durations, buffer times, and stylist skill associations.
- **Scheduling Rules:** Working hours, breaks, holidays, and temporary closures defining available booking windows.
- **Optimization Focus:** Range index optimization (GiST / B-tree) for overlap checks (`tsrange`) and atomic slot locking.

## 6. SAFEGUARDING BUSINESS RULES UNDER PEAK CONCURRENCY

Ensuring Zero Data Inconsistencies or Workflow Violations

### No Double Bookings

INTEGRITY

Utilizing PostgreSQL exclusion constraints (`EXCLUDE USING gist`) or short-lived advisory locks during checkout to guarantee that two simultaneous bookings cannot reserve the same stylist slot.

### No Duplicate Tickets

IDEMPOTENCY

Enforcing client-generated idempotency keys and atomic upserts on orders to prevent staff phones from generating duplicate kitchen tickets during transient connection retries.

### Fast Floor Checkouts

THROUGHPUT

Isolating payment reconciliation writes from real-time ticket display reads, ensuring checkout screens never experience UI freezes during peak dinner ordering rushes.

# Step-by-Step Optimization Methodology

Two-Phase Execution Plan: Telemetry, Synthetic Staging Testing, and Fix Verification

**Phase 1:** Audit, Telemetry & Staging Simulation
**Phase 2:** Remediation, Fixes & Verification
**Tooling:** k6, pg_stat_statements, PgBouncer, EXPLAIN ANALYZE

## 7. PHASE 1: BASELINE AUDIT & SYNTHETIC STAGING LOAD TESTING

Weeks 1–2: Quantifying Current Limits and Replicating Peak Failure Modes

### Step 1.1: Telemetry & Baseline Metrics

TELEMETRY

We establish granular baseline metrics on current database and API performance:

- **pg_stat_statements:** Query frequency, total runtime, mean exec time, and buffer cache misses.
- **Lock Contention:** Profiling transaction wait times, blocked queries, and deadlocks.
- **Connection Utilization:** Monitoring active, idle, and waiting connection pools in PgBouncer.
- **Traffic Profiling:** Tracing automatic page refreshes and Supabase Realtime broadcast payloads.

### Step 1.2: Synthetic Data & Peak Simulation

STAGING HARNESS

We build automated, repeatable load-testing scripts (using k6 / Locust):

- **Synthetic Data Generation:** Seeding staging with 100 realistic restaurants and 100 hair salons with full relational history.
- **Scenario A (Restaurants):** Simulating 100 checkouts + 200 staff mobile phones during a dinner rush.
- **Scenario B (Salons):** Simulating active calendar views, availability lookups, and concurrent bookings.
- **Scenario C (Combined Peak):** Running both workloads simultaneously to surface cross-tenant resource contention.

## 8. PHASE 2: FIX IMPLEMENTATION & REMEDIATION

Weeks 3–4: Deploying Targeted Fixes and Re-Verifying Under Peak Load

### Step 2.1: Query, Index & Lock Tuning

ENGINE TUNING

Applying targeted database optimizations based on empirical trace data:

- **Strategic Indexing:** Composite indexes for multi-column filters, partial indexes for active checks and upcoming bookings, and covering indexes for foreign keys.
- **RLS Refactoring:** Converting slow dynamic table joins in Row-Level Security policies into cached JWT claims and `STABLE` security-definer functions.
- **Lock Optimization:** Restructuring updates on `orders` and `appointments` to eliminate blocking locks and prevent deadlocks.
- **Connection Pooling:** Fine-tuning PgBouncer transaction-mode settings and client connection budgets.

### Step 2.2: Application & Realtime Triage

NEXT.JS & API

Refining the data access layer in Next.js and Supabase Realtime:

- **Channel Scoping:** Filtering Realtime subscriptions by `location_id` so clients only receive events for their venue.
- **Debouncing & Batching:** Replacing aggressive client polling on staff phones with debounced invalidations and batched state syncs.
- **Selective Fetching:** Replacing over-fetching queries with minimal column projections to reduce serialization overhead.

## 9. STEP 2.3: RE-VERIFICATION & COMPARATIVE BENCHMARKING

Proving Measurable Gains Under Identical Staging Conditions

### Objective Verification Protocol

DEFINITIVE PROOF

We re-execute the identical k6 load scripts against the optimized database under the exact same simulated dinner rush and salon booking peak workloads. We verify that:

- Response times for checkouts, kitchen ticket submissions, and slot bookings drop significantly across p50, p95, and p99 percentiles.
- Peak connection counts remain stable within PgBouncer limits with zero connection pool timeouts.
- Zero missing or duplicate orders, and all competing booking requests strictly adhere to salon availability rules.

# Deliverables, Handover & Engineering Team

Comprehensive Test Harness Handover, Capacity Guidance, and Team Credentials

**Engagement:** Remote Technical Specialist
**Availability:** Immediate Kickoff (Within 24–48 Hours)
**Timeline:** 2 Weeks Phase 1 + 2 Weeks Phase 2

## 10. CONCRETE HANDOVER DELIVERABLES

What YouZen Receives at the Conclusion of the Engagement

| DELIVERABLE | DESCRIPTION & CONTENTS | FORMAT |
|---|---|---|
| **Automated Test Suite** | Repeatable, parameterized k6 / Locust load scripts simulating 100 restaurants, 100 salons, and combined peak workloads with synthetic data seeding utilities. | Git repository, CLI scripts, and README execution guide. |
| **Before/After Benchmark Report** | Comprehensive empirical comparison of query response times (p50, p95, p99), request throughput (RPS), peak connection count, lock wait times, and error rates. | Structured Technical Report (PDF / Markdown) with graphs. |
| **Database & RLS Fixes** | All SQL migration scripts for indexes, constraints, rewritten views, optimized RLS policies, and Next.js / Realtime connection code enhancements. | Versioned Supabase SQL migration files and Git PRs. |
| **Capacity & Rollout Runbook** | Long-term platform capacity assessment, recommended compute tier thresholds, production zero-downtime rollout plan, and alerting triggers. | Operational Runbook & Monitoring Thresholds document. |

## 11. DEDICATED SENIOR ENGINEERING TEAM

Full Resumes Attached in Proposal Package

**Viranga Lakshan** — Lead Solutions Architect & Database Eng.

**Role for YouZen:** Primary technical lead. Personally leads the PostgreSQL / Supabase audit, query profiling, lock diagnosis, RLS optimization, and personally implements all agreed database fixes.

**Janith Kavishka** — Backend/API & Load-Testing Eng.

**Role for YouZen:** Primary load-testing lead. Develops reusable k6 / Locust scripts, prepares synthetic data generators for 100 restaurants and 100 salons, and runs peak concurrency simulations.

**Lithumitha Ravikumar** — Project Coordination & Client Comms

**Role for YouZen:** Primary project coordinator. Manages delivery timelines, milestone tracking, client communications, technical runbooks, and delivery sign-offs.

### Immediate Kickoff & Collaboration Plan

We are prepared to begin immediately upon selection. Our first step is a 30-minute technical kickoff call with Romain and the engineering team to review repository access, staging database provisioning, and telemetry capture tools. Phase 1 findings and the initial staging test harness will be delivered within the first two weeks.